

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403573
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Pfeiffer; Michael W.

Multi-tool

Abstract

A multi-tool includes a pair of plates having respective handle portions, joint portions, and head portions, with the plates pivotally connected to each other at the joint portions, and to be pivoted relative to each other to establish an open position and a closed position of the plates. A bit holder is extended from an end of one of the plates, with the bit holder including a socket to receive an insert bit, the handle portion of the one of the plates including a storage area to receive and store the insert bit, and, in the closed position of the plates, the plates to overlap each other with the handle portion of the other of the plates to cover the storage area and the handle portion of the one of the plates.

Inventors:	Pfeiffer; Michael W. (Eden Prairie, MN)
Applicant:	Wolf Tooth Components, LLC (Burnsville, MN)
Family ID:	85385548
Assignee:	Wolf Tooth Components, LLC (Burnsville, MN)
Appl. No.:	17/989511
Filed:	November 17, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230072405 A1	Mar. 09, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 16351484 20190312 US 11623329 child-doc US 17989511
us-provisional-application US 63280616 20211117
us-provisional-application US 62641585 20180312

Publication Classification

Int. Cl.: B25B27/00 (20060101); B25B23/00 (20060101); B25G1/08 (20060101)

U.S. Cl.:

CPC B25B27/0071 (20130101); B25B23/0035 (20130101); B25G1/085 (20130101);

Field of Classification Search

CPC: B25F (1/02); B25F (1/003); B25F (1/04); B25B (7/22); B25B (27/0071); B25B (27/22); B25B (27/24); B25G (1/08); B25G (1/085)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1510953	12/1923	Murphy	N/A	N/A
D150222	12/1947	Gile	N/A	N/A
4336652	12/1981	Robertson	N/A	N/A
D288893	12/1986	Epstein	N/A	N/A
5193336	12/1992	King	N/A	N/A
5255438	12/1992	Morgan	N/A	N/A
5280659	12/1993	Park	7/128	B25F 1/003
D358536	12/1994	Gore	N/A	N/A
5421225	12/1994	Chen	81/177.4	B25F 1/02
5432968	12/1994	Beck	7/128	B25F 1/006
5564318	12/1995	Pail	N/A	N/A
5910174	12/1998	Finn	81/438	B25B 7/00
D416178	12/1998	Moore	N/A	N/A
D424900	12/1999	Budrow et al.	N/A	N/A
6119561	12/1999	Anderson	81/177.4	B25F 1/003
6185771	12/2000	Trusty, Sr.	N/A	N/A
D442451	12/2000	Berg	N/A	N/A
6223373	12/2000	Yeh	7/137	B25B 23/005
D445655	12/2000	Maddox	N/A	N/A
D448637	12/2000	Yeh	N/A	N/A
6305254	12/2000	Hsieh	81/177.4	B25B 13/04
6804869	12/2003	Tong	N/A	N/A
7162758	12/2006	Skinner	7/164	G01C 9/26
7334502	12/2007	Durkee	81/20	B25G 1/085
7346986	12/2007	Feith	N/A	N/A
7587963	12/2008	Broadus	N/A	N/A
D613138	12/2009	Salvino	N/A	N/A
7845254	12/2009	Lionel	81/414	B25F 1/02
8176814	12/2011	Bernstein et al.	N/A	N/A
D666883	12/2011	Howard	N/A	N/A
D689351	12/2012	Webb, Jr.	N/A	N/A
9225153	12/2014	Anderson	N/A	H02G 1/1214

D747638	12/2015	Hernandez	N/A	N/A
D752402	12/2015	Goldy et al.	N/A	N/A
D768460	12/2015	Adelman	N/A	N/A
9457460	12/2015	Hermansen et al.	N/A	N/A
D772028	12/2015	Tangreen	N/A	N/A
10035248	12/2017	Ichikawa	N/A	N/A
D832673	12/2017	Berman	N/A	N/A
D837619	12/2018	George	N/A	N/A
D842067	12/2018	Young	N/A	N/A
D848808	12/2018	Zhong	N/A	N/A
10464199	12/2018	Hardinge	N/A	B25F 1/04
D878890	12/2019	Riehl	N/A	N/A
D886564	12/2019	Dittmer et al.	N/A	N/A
10850382	12/2019	Ragner	N/A	B25F 1/003
11794313	12/2022	Bessac	N/A	B25B 7/10
D1007266	12/2022	Fu	N/A	N/A
2003/0154552	12/2002	Oka	7/127	B25B 7/10
2007/0122574	12/2006	Brown	428/34.1	B25F 1/02
2009/0211034	12/2008	Lionel	7/127	B25F 1/02
2010/0293721	12/2009	James	N/A	N/A
2012/0118115	12/2011	Huang	N/A	N/A
2013/0097786	12/2012	Hardinge	30/162	B25F 1/04
2016/0082793	12/2015	Davidson et al.	N/A	N/A
2017/0157757	12/2016	Winefordner et al.	N/A	N/A
2017/0266803	12/2016	Hardinge	N/A	B25G 3/18
2017/0348830	12/2016	Lazenby	N/A	B26B 17/00
2018/0093369	12/2017	Ichikawa	N/A	N/A
2019/0061126	12/2018	Park et al.	N/A	N/A
2019/0275651	12/2018	Dittmer et al.	N/A	N/A
2020/0171642	12/2019	Newman	N/A	B25F 1/04
2020/0191240	12/2019	Chuang	N/A	N/A
2020/0306935	12/2019	Bessac	N/A	B25F 1/003
2023/0072405	12/2022	Pfeiffer	N/A	N/A
2023/0364813	12/2022	Liang et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
M485775	12/2013	TW	N/A
201531384	12/2014	TW	N/A
M505986	12/2014	TW	N/A
201634194	12/2015	TW	N/A
M536595	12/2016	TW	N/A
M545025	12/2016	TW	N/A
M545057	12/2016	TW	N/A
M546272	12/2016	TW	N/A
M546273	12/2016	TW	N/A
M549279	12/2016	TW	N/A
M549280	12/2016	TW	N/A
M554399	12/2017	TW	N/A
M554400	12/2017	TW	N/A

OTHER PUBLICATIONS

BiB Creative Concept Store—Clever Tool (3 sets), retrieved Mar. 22, 2018, 2 pages, <<http://bibcreative.bigcartel.com/product/clever-tool-3-sets>>. cited by applicant

Bikerumor Clever Standard Makes Quick Work of Re-using Chain Quick Links, Nov. 28, 2016, 3 pages, <<https://www.bikerumor.com/2016/11/28/clever-standard-makes-quick-work-re-using-chain-quick-links/>>. cited by applicant

KMC MissingLink Lever, retrieved Mar. 22, 2018, 1 page, <<https://store.kmcchain.us/p/missinglink-lever>>. cited by applicant

Moosetreds 2 in 1 Bike Tire Levers + Master Link Chain Pliers, retrieved Mar. 22, 2018, 10 pages, <<https://www.amazon.com/Moosetreds-Levers-Premium-Engineering-Plastic/dp/B074QM8NK2>>. cited by applicant

Park Tool Master Link Pliers MLP-1.2, retrieved Mar. 22, 2018, 2 pages, <<https://www.parktool.com/product/master-link-pliers-mlp-1-2>>. cited by applicant

Sumnacon Bike Tire Levers + Master Link Chain Pliers—Sumnacon 3 in 1 Multifunction Bicycle Tire Repair Tool, retrieved Mar. 22, 2018, 11 pages, <https://www.amazon.com/Bike-Levers-Master-Chain-Pliers/dp/B078HK9NPP/ref=sr_1_17?s=spo>. cited by applicant

LEZYNE Multi-Chain Bicycle Pliers, Apr. 29, 2020 [online], [retrieved Jul. 24, 2024]. Retrieved from internet, https://www.amazon.com/dp/B087WQ7CZZ/ref=sspa_dk_detail_1?pf_rd_p=953c7d66-4120-4d22-a777- (see last page of attached pdf for full URL) (Year: 2020). cited by applicant

8-Bit Pack Pliers, Nov. 17, 2020 [online], [retrieved Jul. 24, 2024]. Retrieved from internet, <https://www.wolftoothcomponents.com/collections/tools/products/8-bitpack-pliers> (Year: 2020). cited by applicant

Multi-Functional Master Link Pliers with Master Link Removing, May 27, 2023 [online], [retrieved Jul. 24, 2024]. Retrieved from internet, https://www.amazon.com/KIEVODE-Multi-Functional-Master-Removing-Storage/dp/B0C6GP5249/ref=sr_1_14? (see last page of attached pdf for full URL) (Year: 2023). cited by applicant

Primary Examiner: Thomas; David B.

Attorney, Agent or Firm: Dicke, Billig & Czaja, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation-in-Part of U.S. patent application Ser. No. 16/351,484 filed on Mar. 12, 2019, which claims priority under 35 U.S.C. 119(e) to U.S. Provisional Patent Application Ser. No. 62/641,585 filed on Mar. 12, 2018, and this application claims priority under 35 U.S.C. 119(e) to U.S. Provisional Patent Application Ser. No. 63/280,616 filed on Nov. 17, 2021, each of which is incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates generally to a multi-tool and, more specifically, relates to a multi-tool for use with a bicycle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1** and **2** are front and rear views, respectively, of an example of a multi-tool.
- (2) FIGS. **3** and **4** are exploded front perspective views of an example of the multi-tool of FIGS. **1** and **2**.
- (3) FIGS. **5** and **6** are exploded rear perspective views of an example of the multi-tool of FIGS. **1** and **2**.
- (4) FIGS. **7** and **8, 9** are front and front perspective views, respectively, of the multi-tool of FIGS. **1** and **2** in a closed position, with a bit holder of the multi-tool in a first position.
- (5) FIGS. **10** and **11, 12** are rear and rear perspective views, respectively, of the multi-tool of FIGS. **1** and **2** in a closed position.
- (6) FIGS. **13** and **14** are side views of the multi-tool of FIGS. **1** and **2** in a closed position.
- (7) FIGS. **15** and **16** are front perspective views of the multi-tool of FIGS. **1** and **2**, with the bit holder of the multi-tool in different positions.
- (8) FIGS. **17** and **18** are front and rear perspective views, respectively, of the multi-tool of FIGS. **1** and **2** in an open position.
- (9) FIG. **19** is a front view of the multi-tool of FIGS. **1** and **2** in an example of a position between the open position of FIG. **17** and the closed position of FIG. **7**.
- (10) FIGS. **20A, 20B, 20C** are front perspective views illustrating an example of use of the multi-tool of FIGS. **1** and **2**.
- (11) FIGS. **21A, 21B** and FIGS. **22A, 22B** are front and rear perspective views, respectively, illustrating an example of use of the multi-tool of FIGS. **1** and **2**.
- (12) FIG. **23** is a front perspective view of an example of the multi-tool of FIGS. **1** and **2**, with a rasp, set of chain links, plurality of insert bits, and a wrench.
- (13) FIG. **24** is a front view illustrating an example of the multi-tool of FIGS. **1** and **2**.
- (14) FIGS. **25A, 25B, 25C** are cross-sectional views of a portion of the multi-tool of FIGS. **1** and **2** from the perspective of line A-A of FIG. **24**.
- (15) FIGS. **26A, 26B, 26C** are cross-sectional views of a portion of the multi-tool of FIGS. **1** and **2** from the perspective of line B-B of FIG. **24**.
- (16) FIGS. **27, 28, 29, 30, 31** illustrate examples of use of the multi-tool of FIGS. **1** and **2**.
- (17) FIG. **32** includes views of an example of a rasp for use with the multi-tool of FIGS. **1** and **2**.
- (18) FIGS. **33A, 33B** illustrate an example of use of the rasp of FIG. **32**.

DETAILED DESCRIPTION

(19) In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific examples in which the disclosure may be practiced. It is to be understood that other examples may be utilized and structural or logical changes may be made without departing from the scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims.

- (20) FIGS. **1** and **2** are front and rear views, respectively, of an example of a multi-tool **10**. In one example, multi-tool **10** includes handles **12**, joint **14**, head **16**, and bit holder **18**. As disclosed herein, handles **12** are pivotally connected at joint **14**. As such, head **16** may be opened and/or closed (or expanded and/or retracted) by actuation of handles **12**. In the illustrated example, multi-tool **10** is formed by a pair of plates **100** and **200**, with plates **100** and **200** having respective handle portions **110** and **210**, joint portions **120** and **220**, and head portions **130** and **230**. In addition, bit holder **18** is pivotally or rotatably coupled with one of the plates (for example, plate **200**) such that bit holder **18** may be swiveled, pivoted or rotated between different positions, as disclosed herein.
- (21) FIGS. **3** and **4** are exploded front perspective views of multi-tool **10**, and FIGS. **5** and **6** are exploded rear perspective views of multi-tool **10**. In one implementation, plate **100** has opposite sides **101** and **102**, and opposite edges **103** and **104**, and plate **200** has opposite sides **201** and **202**,

and opposite edges **203** and **204**. In one implementation, edges **103** and **104** and edges **203** and **204** extend along respective handle portions **110** and **210** and respective joint portions **120** and **220** of respective plates **100** and **200**.

(22) As disclosed herein, plates **100** and **200** are pivotally connected such that sides **102** and **201** of respective plates **100** and **200** oppose or face each other in a closed position. In one example, in the closed position, plates **100** and **200** nest or overlap with each other. As such, edges **103** and **203** of respective plates **100** and **200** are substantially aligned, and edges **104** and **204** of respective plates **100** and **200** are substantially aligned (see, e.g., FIG. 7). In one example, handle portions **110** and **210** of respective plates **100** and **200** are each of a generally rectangular shape and, in one implementation, include beveled and/or rounded edges.

(23) In one implementation, a thickness **T1** of handle portion **110** of plate **100** is less than a thickness **T2** of joint portion **120** of plate **100** (see, e.g., FIG. 13) such that a shoulder **106** is formed on side **102** of plate **100**. In addition, in one implementation, a thickness **T3** of joint portion **220** of plate **200** is less than a thickness **T4** of handle portion **210** of plate **200** (see, e.g., FIG. 14) such that a shoulder **206** is formed on side **201** of plate **200**.

(24) In the illustrated example, joint **14** includes a bolt **20** and a post **22**. In one implementation, joint portion **120** of plate **100** has a hole **122** formed therethrough, and post **22** is formed on joint portion **220** of plate **200**. In one implementation, post **22** extends into and/or passes through hole **122** such that plate **100** is rotatably supported by (or rotatably supported on) post **22**. In addition, in one implementation, post **22** includes internal threads **222**. As such, in one implementation, bolt **20** extends into and/or passes through hole **122**, and is threaded into post **22** to pivotally couple plates **100** and **200**. In one implementation, bolt **20** is a chainring bolt. Bolt **20**, therefore, may be used to replace a chainring bolt on a bicycle.

(25) In one implementation, joint **14** includes a washer **24** interposed between joint portions **120** and **220** of respective plates **100** and **200**. As such, washer **24** helps to space apart or position plates **100** and **200** relative to each other and provides a bearing surface for pivotal movement of plates **100** and **200** relative to each other. Washer **24** is supported by (or on) post **22** and, in one implementation, is fit within a recess **124** formed in side **012** of plate **100**. In one example, washer **24** is formed of a polymer material, including, for example, an acetal, such as Delrin, or a polyamide, such as Nylon.

(26) In the illustrated example, head portions **130** and **230** extend from an end of respective plates **100** and **200**. In one implementation, head portions **130** and **230** each have respective jaw sections **132** and **232** and respective tips **134** and **234**. Jaw sections **132** and **232** oppose each other and, as disclosed herein, together form gripping jaws **17** of multi-tool **10** (see, e.g., FIG. 19). In one implementation, jaw sections **132** and **232** include respective and opposing gripping surfaces **133** and **233** (see also FIG. 19). In one example, gripping surfaces **133** and **233** each have a concave shape, including, for example, a concave arcuate shape, a U-shape, or a V- or shallow V-shape.

(27) In one example, tips **134** and **234** are configured to disengage and/or engage or remove and/or install a chain link of a roller chain, such as a master link for a roller chain of a bicycle, including, for example, a Quick-Link by Shimano. More specifically, as disclosed herein, tips **134** and **234** are sized and shaped such that tips **134** and **234** may engage teeth receiving openings formed by spaced link plates of a roller chain. In one implementation, tips **134** and **234** are sized and shaped as single or individual teeth. As such, tips **134** and **234** have respective ends **135** and **235** and respective faces or engaging surfaces **136**, **137** and **236**, **237** (see also FIG. 19). In one example, engaging surfaces **136**, **137** and **236**, **237** each have a concave shape, including, for example, a concave arcuate shape.

(28) In the illustrated example, head portions **130** and **230** of respective plates **100** and **200** include respective shoulders **138** and **238**. In one implementation, as disclosed herein, shoulders **138** and **238** limit or stop rotation of plates **100** and **200** relative to each other.

(29) In one example, handles **12** are magnetically attracted to each other. More specifically, in one

implementation, handle portions **110** and **210** include respective magnetic elements **31** and **32**. As such, as disclosed herein, magnetic elements **31** and **32** provide a magnetic closure to and magnetically couple handle portions **110** and **210** of respective plates **100** and **200**.

(30) In one example, magnetic elements **31** and **32** are attached to, secured to, or positioned on or in opposing sides of respective handle portions **110** and **210**. For example, magnetic element **31** of handle portion **110** is attached to, secured to, or positioned on or in side **102** of plate **100**, and magnetic element **32** of handle portion **210** is attached to, secured to, or positioned on or in side **201** of plate **200**. In the illustrated example, magnetic elements **31** and **32** are attached to, secured to, or positioned on or in opposing sides of respective handle portions **110** and **210** within respective recesses or cavities **112** and **212**.

(31) In one example, magnetic element **32** is or includes a magnet, and magnetic element **31** is or includes a magnet or is or includes an element (i.e., keeper) formed of a material to which a magnet may be attracted (e.g., a ferrous material). Magnetic elements **31** and **32** each may be a disk, plate, strip, or sheet, and each may be of a circular, rectangular, or other shape.

(32) In the illustrated example, bit holder **18** is pivotally or rotatably coupled with plate **200**. More specifically, in one implementation, bit holder **18** is pivotally or rotatably coupled with plate **200** such that bit holder **18** is extended from an end **219** of handle portion **210** of plate **200**. In one example, bit holder **18** is pivotally or rotatably coupled with plate **200** by a pin **26**. In examples, bit holder **18** is pivotally or rotatably coupled with plate **200** such that bit holder **18** may be swiveled, pivoted or rotated between different positions. In examples, bit holder **18** may be swiveled, pivoted or rotated in multiple directions to establish multiple positions of bit holder **18**. In one implementation, a detent **28** may be provided at (adjacent or near) end **219** of plate **200** to help maintain a position of bit holder **18** as bit holder **18** is moved or swiveled, pivoted or rotated between different positions. In one example, bit holder **18** may include one or more than groove **181** which may be engaged by detent **28** to establish the different positions of bit holder **18**.

(33) In the illustrated example, bit holder **18** is a hex head and includes a hex socket **30** (FIG. **15**) to receive a hex insert bit, as an example of an insert bit, as disclosed herein. As such, in examples, bit holder **18** may be used as a hex key wrench or as a driver for a hex insert bit (i.e., bit driver). In one implementation, a magnet (not shown) may be provided within bit holder **18** to help hold a respective insert bit within bit holder **18**.

(34) In examples, plate **100** (including handle portion **110** of plate **100**) includes a storage area **140**. In one implementation, storage area **140** is configured to receive and store a rasp, as disclosed herein. In the illustrated example, storage area **140** includes a recessed region **142** formed in side **102** of plate **100** and a slot **144** formed in plate **100** within recessed region **142**. In addition, storage area **140** includes a pocket **146** at an end of recessed region **142** and a crossbar **148** across slot **144**.

(35) In one example, a magnetic element **33** is provided within recessed region **142** to help retain or hold a rasp within recessed region **142** and slot **144**, as disclosed herein. In the illustrated example, magnetic element **33** is attached to, secured to, or positioned on or in side **102** of plate **100** within a recess or cavity **114**. In one example, magnetic element **33** is or includes a magnet, and may be a disk, plate, strip, or sheet, and may be of a circular, rectangular, or other shape.

(36) In examples, plate **200** (including handle portion **210** of plate **200**) includes one or more than one storage area. In one implementation, plate **200** (including handle portion **210** of plate **200**) includes a storage area **240**, a storage area **250**, and a storage area **260**.

(37) In one implementation, storage area **240** is configured to receive and store a chain link for a roller chain, such as a master link for a roller chain of a bicycle, including, for example, a Quick-Link by Shimano, as disclosed herein. In the illustrated example, storage area **240** includes a recessed region **242** formed in side **201** of plate **200** and one or more than one hole **244** formed through plate **200** within recessed region **242**. As disclosed herein, recessed region **242** is configured to receive and store a link plate of a roller chain and hole **244** is configured to receive a link pin extended from the link plate.

(38) In one example, recessed region **242** is configured to receive and store multiple chain links. In the illustrated example, recessed region **242** is configured to receive and store two chain links (i.e., one pair (or one set) of mating chain link plates), and includes two holes **244**, each to receive a respective link pin. In one example, holes **244** communicate with a common recess **246** at side **202** of plate **200**. In other examples, each hole **244** may include a corresponding recess **246**, such as a countersink, counterbore, spot face or other recess, at side **202** of plate **200**.

(39) In one example, a magnetic element **34** is provided within recessed region **242** to help retain or hold a chain link within recessed region **242** of storage area **240**, as disclosed herein. In the illustrated example, magnetic element **34** is attached to, secured to, or positioned on or in side **201** of plate **200** within a recess or cavity **214**. In one example, magnetic element **34** is or includes a magnet, and may be a disk, plate, strip, or sheet, and may be of a circular, rectangular, or other shape.

(40) In one implementation, storage area **250** is configured to receive and store a plurality of insert bits, as disclosed herein. More specifically, in the illustrated example, storage area **250** includes a plurality of cavities or pockets **252** for holding or storing insert bits. In one implementation, pockets **252** are each sized to receive a respective insert bit. In the illustrated example, pockets **252** are contiguous and include two rows of three pockets. As such, in the illustrated example, a total of six insert bits may be stored in pockets **252**. Although two rows of three pockets each are illustrated and described, the number and/or configuration or arrangement of pockets **252** may vary.

(41) In the illustrated example, an opening **254** is provided in plate **200** within storage area **250**. In one implementation, opening **254** communicates with one or more than one pocket **252** such that an insert bit positioned within the respective pocket **252** may be accessed through opening **254**. As such, a respective insert bit may be more easily removed from a respective pocket **252**, as disclosed herein.

(42) In one example, one or more than one magnetic element **36** is provided within storage area **250** to help hold or retain insert bits within respective pockets **252**. In one implementation, magnetic element **36** is positioned within a recesses or cavity **256** communicated with one or more than one pocket **252**. In the illustrated example, cavity **256** extends along multiple pockets **252** such that one magnetic element **36** extends along multiple pockets **252**. In one implementation, magnetic elements **36** are positioned within respective recesses or cavities **256** provided in first and second rows of pockets **252**. As such, in one implementation, two magnetic elements **36** are used for six pockets **252**. Although illustrated as being oblong in shape, magnetic elements **36** may be of a circular, rectangular, or other shape, and may be a disk, plate, strip, or sheet.

(43) In one implementation, storage area **260** is configured to receive and store a wrench. More specifically, in one example, storage area **260** is configured to receive and store a combination spoke wrench and valve core wrench, as disclosed herein. In the illustrated example, storage area **260** includes a recessed region **262** formed in side **202** of plate **200**. In one example, magnetic element **34** is positioned opposite recessed region **262** such that magnetic element **34** helps to retain or hold a wrench within recessed region **262** of storage area **260**, as disclosed herein.

(44) FIGS. **7** and **8**, **9** are front and front perspective views, respectively, of multi-tool **10** in a closed position, and FIGS. **10** and **11**, **12** are rear and rear perspective views, respectively, of multi-tool **10** in a closed position. In addition, FIGS. **13** and **14** are side views of multi-tool **10** in a closed position.

(45) In the illustrated example, in the closed position, plates **100** and **200** nest or overlap with each other. More specifically, in the closed position, edges **103** and **203** and edges **104** and **204** of respective plates **100** and **200**, as extended along respective handle portions **110** and **210** and extended along respective joint portions **120** and **220**, are substantially aligned with each other.

(46) In one implementation, tips **134** and **234** of respective head portions **130** and **230** are oriented at an angle relative to respective plates **100** and **200**. More specifically, in one example, axes **139** and **239**, extended through ends **135** and **235** of respective tips **134** and **234**, are oriented at

respective angles **A1** and **A2** relative to edges **103** and **204** of respective plates **100** and **200**.

(47) In one implementation, plates **100** and **200** are pivotally coupled at joint **14** such that joint portions **120** and **220** overlap with each other. More specifically, in one example, shoulders **106** and **206** are correspondingly positioned on sides **102** and **201** of respective plates **100** and **200**, and plates **100** and **200** are pivotally coupled, such that, in a closed position of multi-tool **10**, joint portions **120** and **220** nest with each other.

(48) In the illustrated example, in the closed position, plate **200** extends beyond plate **100**. More specifically, end **219** of handle portion **210** of plate **200** extends beyond an end **119** of handle portion **110** of plate **100**. As such, in one example, in the closed position, bit holder **18** extends beyond end **119** of plate **100**.

(49) FIGS. **15** and **16** are front perspective views of multi-tool **10**, with bit holder **18** in different positions. In examples, bit holder **18** may be swiveled, pivoted or rotated between a first position (as illustrated, for example, in FIG. **8**) and one or more than one second position (as illustrated, for example, in FIGS. **15**, **16**). More specifically, in one implementation, bit holder **18** may be swiveled, pivoted or rotated in two directions (for example, to the left and to the right) to establish two second positions of bit holder **18**. In examples, the first position of bit holder **18** represents a neutral position of bit holder **18**. In examples, the first position and the second position(s) of bit holder **18** represent operable positions of bit holder **18**, as disclosed herein.

(50) In one example, in the first position, bit holder **18** is aligned with plate **200** such that an axis **19** of bit holder **18** is parallel (substantially) with an axis **209** of plate **200**. In examples, in the second position, bit holder **18** is oriented at an angle to plate **200** such that axis **19** of bit holder **18** is oriented at an angle to axis **209** of plate **200**. As illustrated in the examples of FIGS. **15**, **16**, bit holder **18** is oriented at an angle to plate **200** such that axis **19** of bit holder **18** is perpendicular (substantially) to axis **209** of plate **200**.

(51) FIGS. **17** and **18** are front and rear perspective views, respectively, of multi-tool **10** in an open position. In the illustrated example, from a closed position (e.g., FIG. **7**, **8**) to an open position (e.g., FIG. **17**, **18**), handle portion **110** of plate **100** is rotated or pivoted relative to handle portion **210** of plate **200** about joint **14**. More specifically, handle portion **110** of plate **100** is pivoted, for example, away from handle portion **210** of plate **200**, as represented by arrow **R1**. As such, jaw section **132** of head portion **130** is pivoted relative to jaw section **232** of head portion **230**. More specifically, jaw section **132** of head portion **130** is pivoted, for example, away from jaw section **232** of head portion **230**, as represented by arrow **R2**. Thus, in the illustrated example, gripping jaws **17** are in an example of an open position. In an open position, gripping jaws **17** may be used to grip or hold an object, for example, a bolt, nut, or other object.

(52) In the illustrated example, handle portion **110** of plate **100** is pivoted relative to handle portion **210** of plate **200** by approximately 90 degrees of rotation. As such, jaw section **132** of head portion **130** is pivoted relative to jaw section **232** of head portion **230**, by approximately 90 degrees of rotation. In other examples, handle portion **110** of plate **100** may be pivoted relative to handle portion **210** of plate **200** by other degrees of rotation including, for example, degrees of rotation between approximately 0 degrees and approximately 90 degrees. In addition, in examples, handle portion **110** of plate **100** may be pivoted relative to handle portion **210** of plate **200** by greater than 90 degrees of rotation. Although handle portion **110** of plate **100** is illustrated as being pivoted relative to handle portion **210** of plate **200**, handle portion **210** of plate **200** may be pivoted relative to handle portion **110** of plate **100**, or handle portion **110** of plate **100** and handle portion **210** of plate **200** may both be pivoted relative to each other.

(53) In one implementation, shoulders **138** and **238** of respective head portions **130** and **230** provide stops to limit pivoting or rotation of plates **100** and **200** relative to each other. For example, in the illustrated example, edge **104** of plate **100** engages or contacts shoulder **238** of plate **200** and/or edge **203** of plate **200** engages or contacts shoulder **138** of plate **100** such that rotation of plates **100** and **200** relative to each other is limited or stopped.

(54) FIG. 19 is a front view of multi-tool 10 in an example of a position between the open position of FIG. 17 and the closed position of FIG. 7. In the illustrated example, from the open position of FIG. 17, handle portion 110 of plate 100 is pivoted relative to handle portion 210 of plate 200, for example, toward handle portion 210 of plate 200, as represented by arrow R3. As such, jaw section 132 of head portion 130 of plate 100 is pivoted relative to jaw section 232 of the head portion 230 of plate 200, for example, toward jaw section 232, as represented by arrow R4. Thus, in the illustrated example, gripping jaws 17 are in another example of an open position. As such, gripping jaws 17 may be used to grip or hold an object, for example, a bolt, nut or other object.

(55) In one example, as handle portion 110 of plate 100 and handle portion 210 of plate 200 are brought into proximity of each other, for example, as handle portion 110 of plate 100 is pivoted relative to handle portion 210 of plate 200 and toward the closed position of FIG. 7, magnetic elements 31 (see, e.g., FIG. 18) and 32 interact or attract each other to magnetically “pull” handle portion 110 of plate 100 to the closed position of FIG. 7. As such, in one example, magnetic elements 31 and 32 provide a magnetic closure and magnetically couple handle portions 110 and 210 of respective plates 100 and 200. In one example, magnetic elements 31 and 32 help to maintain the closed position of multi-tool 10 until an open position of multi-tool 10 is initiated (e.g., until handle portion 110 of plate 100 is pivoted or rotated away from handle portion 210 of plate 200, as illustrated, for example, in FIG. 17).

(56) FIGS. 20A, 20B, 20C are front perspective views, respectively, illustrating an example of use of multi-tool 10. For example, multi-tool 10 may be used to disengage and/or engage or remove and/or install a chain link 50 of a roller chain 60, such as a master link for a roller chain of a bicycle, including, for example, a Quick-Link by Shimano. In one example, chain link 50 may be used to join or couple two ends of roller chain 60 so as to form an endless or continuous loop of roller chain 60.

(57) In the illustrated example, roller chain 60 includes and/or is formed by alternating, overlapping pairs of laterally spaced link plates 62. In one example, alternating, overlapping pairs of spaced link plates 62 are pivotally interconnected by link pins 64 extended through cylindrical rollers 66 provided between spaced link plates 62 at overlapping, opposite ends of spaced link plates 62. As such, laterally spaced link plates 62 form alternating teeth receiving openings 68.

(58) As illustrated in the example of FIG. 20A, tips 134 and 234 of multi-tool 10 engage a section of roller chain 60. More specifically, tips 134 and 234 of multi-tool 10 engage teeth receiving openings 68 formed by spaced link plates 62 of roller chain 60 at opposite ends of chain link 50.

(59) As such, as illustrated in the example of FIG. 20B, handle portions 110 and 210 are pivoted toward each other (e.g., squeezed together), as represented by arrows R5, to disengage chain link 50, as represented by arrow D1. More specifically, engaging surfaces 137 and 237 (see, e.g., FIG. 19) of respective tips 134 and 234 engage chain link 50 and slide link pins of chain link 50 within link pin slots formed in respective link plates of chain link 50. As such, chain link 50 may be released and/or removed from roller chain 60.

(60) As illustrated in the example of FIG. 20C, handle portions 110 and 210 are pivoted away from each other (e.g., pulled apart), as represented by arrows R6 to engage (or install) chain link 50, as represented by arrow E1. More specifically, engaging surfaces 136 and 236 (see, e.g., FIG. 19) of respective tips 134 and 234 engage cylindrical rollers 66 at opposite ends of chain link 50 and slide link pins of chain link 50 within link pin slots formed in respective link plates of chain link 50. As such, chain link 50 is installed on roller chain 60.

(61) FIGS. 21A, 21B and FIGS. 22A, 22B are front and rear perspective views, respectively, illustrating an example of use of multi-tool 10. For example, multi-tool 10 may be used to store a rasp, may be used to store one or more than one chain link, may be used to store one or more than one insert bit, and/or may be used to store a wrench. More specifically, in examples, storage area 140 of handle portion 110 may be used to store a rasp 40, storage area 240 of handle portion 210 may be used to store a chain link for a roller chain, such as, for example, chain link 50, storage area

250 of handle portion **210** may be used to store a plurality of insert bits **70**, and storage area **260** of handle portion **210** may be used to store a combination spoke wrench and valve core wrench **80**.

(62) As illustrated in the example of FIGS. **21A** and **21B** and FIGS. **22A** and **22B**, rasp **40** is fit within recessed region **142** such that one end of rasp **40** fits within pocket **146** (with crossbar **148** providing support for rasp **40** near or adjacent the one end), and an opposite end of rasp **40** is held by magnetic element **33** such that rasp **40** is stored within storage area **140**. In one implementation, rasp **40** is inserted into pocket **146** and positioned within recessed region **142** from side **102** of plate **100**. In one implementation, rasp **40** may be removed from recessed region **142** by contacting rasp **40** through slot **144** from side **101** of plate **100** and pushing rasp **40** out of recessed region **142** away from side **102**.

(63) In one example, chain link **50** includes a pair of corresponding link plates **51** and **52** having respective link pins **53** and **54** extending therefrom. As such, link pins **53** and **54** engage corresponding link pin slots **56** and **55** formed in respective link plates **52** and **51** to assemble chain link **50**. In examples, recessed region **242** of storage area **240** is sized and shaped to receive link plates **51** and **52** of chain link **50**, and holes **244** of storage area **240** are sized and positioned to receive link pins **53** and **54** extended from respective link plates **51** and **52**.

(64) As illustrated in the example of FIGS. **21A** and **21B**, link plates **51** and **52** fit within recessed region **242** of storage area **240**. In addition, as illustrated in the example of FIGS. **22A** and **22B**, link pins **53** and **54** of respective link plates **51** and **52** engage and/or fit within respective holes **244** of storage area **240**. As such, link plates **51** and **52** of chain link **50** are stored within storage area **240**.

(65) In one implementation, magnetic element **34** is positioned within recessed region **242** of storage area **240** such that link plates **51** and **52** each overlap at least a portion of magnetic element **34** when link plates **51** and **52** are positioned within recessed region **242**. As such, link plates **51** and **52** are attracted by magnetic element **34** such that magnetic element **34** helps to retain or hold link plates **51** and **52** within recessed region **242** of storage area **240**. In addition, in the closed position, plate **100** covers storage area **240** such that plate **100** helps to retain or hold link plates **51** and **52** within recessed region **242** of storage area **240**.

(66) In one implementation, as illustrated in the example of FIGS. **22A** and **22B**, respective ends of link pins **53** and **54** protrude into or through recess **246** at side **202** of plate **200**. As such, recess **246** helps to provide access to link pins **53** and **54** for removal of link plates **51** and **52** from storage area **240**, for example, by pressing (pushing) link pins **53** and **54** from side **202** of plate **200**. In the illustrated example, two chain links **50** (each including a pair of corresponding link plates **51** and **52** having respective link pins **53** and **54** and respective link pin slots **55** and **56**) are stored within storage area **240**. The number of chain links stored within storage area **240**, however, may vary.

(67) As illustrated in the example of FIGS. **21A** and **21B**, insert bits **70** are fit within respective cavities or pockets **252** such that insert bits **70** are stored within storage area **250**. In one implementation, magnetic elements **36**, **36** help to hold or retain insert bits within respective pockets **252** of storage area **250**. In addition, in the closed position, plate **100** covers storage area **250** such that plate **100** helps to retain or hold insert bits **70** within pockets **252** of storage area **250**. In examples, as disclosed herein, insert bits **70** may be removed from respective pockets **252** by tipping and/or picking insert bits **70** from side **201** of plate **200** and/or pushing or tipping insert bits **70** from side **202** of plate **200**. In the illustrated example, six insert bits **70** are stored within pockets **252** of storage area **250**. The number of insert bits stored within storage area **240**, however, may vary.

(68) As illustrated in the example of FIGS. **22A** and **22B**, wrench **80** is fit within recessed region **262** such that wrench **80** is stored within storage area **260**. In one implementation, magnetic element **34** (as positioned within recessed region **242** of storage area **240** (e.g., FIG. **21A**) opposite recessed region **262**) helps to retain or hold wrench **80** within recessed region **262** of storage area **260**.

(69) Accordingly, as illustrated in the example of FIG. 23, multi-tool **10** provides storage for rasp **40**, chain links **50**, insert bits **70**, and/or wrench **80**. In one implementation, insert bits **70** have, for example, the following tips and/or features: Hex: 2 mm, 2.5 mm, 3 mm, 4 mm, 5 mm, 6 mm; Torx® compatible: T10, T25; Flathead #3.5; Phillips #2. In other implementations, insert bits **70** may have other tips and/or features.

(70) In one implementation, as illustrated, for example, in FIGS. 25A, 25B, and 25C, one or more of pockets **252** may include a recess or depression **258** in a bottom (base or floor) of a respective pocket **252** at (adjacent or near) a first end thereof (see also, e.g., FIGS. 3, 4). As such, as illustrated in the examples of FIGS. 25B, 25C, a first end of a respective insert bit **70** may be pushed into depression **258** (as represented, for example, by arrow A1) to tip an opposite, second end of insert bit **70** away from or out of the respective pocket **252** (as represented, for example, by arrow A2). As such, in examples, insert bit **70** may be more easily removed from a respective pocket **252**.

(71) In one implementation, as illustrated, for example, in FIGS. 26A, 26B, 26C, opening **254** of storage area **250** communicates with one or more than one pocket **252** such that an insert bit **70** positioned within the respective pocket **252** may be accessed through opening **254** from side **202** of plate **200**. As such, as illustrated in the examples of FIGS. 26B, 26C, a respective insert bit **70** may be pushed out of a respective pocket **252** (as represented, for example, by arrow A3) such that insert bit **70** is tipped away from or out of the respective pocket **252** (as represented, for example, by arrow A4). As such, in examples, insert bit **70** may be more easily removed from a respective pocket **252**.

(72) FIG. 27 illustrates an example of use of multi-tool **10**. For example, gripping jaws **17** of multi-tool **10** may be used to grip and/or hold an object. In one example, as illustrated in FIG. 27, gripping jaws **17**, as formed by jaw sections **132** and **232**, may be used to grip and/or hold a nut, such as a jam or retaining nut **92** (see also FIG. 31) of a valve stem **94** of an inner tube or a wheel rim, including, for example, an inner tube or a wheel rim of a bicycle. As such, with gripping jaws **17**, jam or retaining nut **92** may be loosened and/or tightened or removed and/or installed.

(73) FIGS. 28 and 29 illustrate examples of use of multi-tool **10**. For example, as illustrated in FIG. 28, bit holder **18** of multi-tool **10** may be used as a driver for an insert bit **70**. In addition, as illustrated in FIG. 29, bit holder **18** of multi-tool **10** may be used as a hex key wrench.

(74) FIGS. 30 and 31 illustrate examples of use of multi-tool **10**. For example, as illustrated in FIG. 30, multi-tool **10** may be used as a spoke wrench. More specifically, combination spoke wrench and valve core wrench **80** may be used as a spoke wrench to engage a spoke or a spoke nipple of a wheel rim, such as, for example, a spoke nipple **98** of a wheel rim of a bicycle. Thus, with combination spoke wrench and valve core wrench **80**, spoke nipple **98** may be loosened and/or tightened and/or removed and/or installed. In addition, as illustrated in FIG. 31, multi-tool **10** may be used as a valve core tool. More specifically, combination spoke wrench and valve core wrench **80** may be used as a valve core tool to engage a valve core of a valve stem of an inner tube or a wheel rim, such as, for example, a valve core **99** of a valve stem an inner tube or a wheel rim of a bicycle. Thus, with combination spoke wrench and valve core wrench **80**, valve core **99** may be loosened and/or tightened and/or removed and/or installed. As illustrated in the examples of FIGS. 30 and 31, a shank of wrench **80** may be inserted into bit holder **18**, operated or driven by multi-tool **10**, as an example of a bit driver.

(75) FIG. 32 includes views of an example of rasp **40**, and FIGS. 33A, 33B illustrate an example of use of rasp **40**. In the illustrated example, rasp **40** is of a slender rod of cylindrical shape with a pointed end **42** and a knurled surface. In one implementation, rasp **40** includes a notched end **44** and a recessed portion **46** such that recessed portion **46** accommodates crossbar **148** and notched end **44** contacts an end of recessed portion **142** when pointed end **42** is inserted into pocket **146** when rasp **40** is stored in storage area **140**. As illustrated in the example of FIGS. 33A, 33B, rasp **40** may be used to clean a valve stem of an inner tube or a wheel rim, such as, for example, a valve stem **94** of an inner tube or a wheel rim of a bicycle.

(76) A multi-tool as disclosed herein includes multiple features and provides multiple uses, including, for example, installing and/or removing a chain link of a roller chain, gripping jaws, bit driver, rasp storage, chain link storage, insert bit storage, and wrench storage.

(77) Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

Claims

1. A multi-tool, comprising: a pair of plates having respective handle portions, joint portions, and head portions, the plates pivotally connected to each other at the joint portions, and to be pivoted relative to each other to establish an open position and a closed position of the plates; and a bit holder extended from an end of one of the plates, the bit holder including a socket to receive an insert bit, the handle portion of the one of the plates including a storage area to receive and store the insert bit, and in the closed position of the plates, the plates to overlap each other with the handle portion of the other of the plates to cover the storage area and opposite edges of the handle portion of the one of the plates.
2. The multi-tool of claim 1, the bit holder to be pivoted between a first position with the bit holder aligned with the one of the plates and a second position with the bit holder oriented at an angle to the one of the plates.
3. The multi-tool of claim 1, the storage area including a pocket formed in a side of the one of the plates, the pocket to receive and store the insert bit.
4. The multi-tool of claim 3, the storage area including a depression formed in the one of the plates within the pocket, one end of the insert bit to be pushed into the depression to tip an opposite end of the insert bit out of the pocket.
5. The multi-tool of claim 3, the storage area including an opening formed through the one of the plates within the pocket, the opening to provide access to the insert bit from an opposite side of the one of the plates.
6. The multi-tool of claim 3, the storage area including a magnetic element to retain the insert bit within the pocket.
7. The multi-tool of claim 3, the insert bit comprising a plurality of insert bits, and the pocket comprising a plurality of pockets, the plurality of pockets to receive and store the plurality of insert bits.
8. The multi-tool of claim 1, the handle portion of the one of the plates including another storage area to receive and store a chain link for a roller chain, in the closed position of the plates, the handle portion of the other of the plates to cover the another storage area.
9. The multi-tool of claim 1, the storage area formed in one side of the one of the plates, and the one of the plates including another storage area formed in an opposite side of the one of the plates to receive and store a wrench.
10. The multi-tool of claim 1, the handle portion of the other of the plates including another storage area to receive and store a rasp, the another storage area including a recessed region formed in the other of the plates to receive and store the rasp.
11. A multi-tool, comprising: a pair of handles; a joint to pivotally connect the handles; a head to be opened and closed by actuation of the handles; and a bit holder extended from an end of one of the handles, the bit holder including a socket to receive an insert bit, the one of the handles including a storage area to receive and store the insert bit, and in a closed position of the handles, the other of the handles to cover the storage area and opposite edges of the one of the handles.

12. The multi-tool of claim 11, the bit holder to be pivoted between a first position with the bit holder aligned with the one of the handles and a second position with the bit holder oriented at an angle to the one of the handles.
 13. The multi-tool of claim 11, the storage area including a pocket to receive and store the insert bit.
 14. The multi-tool of claim 13, the storage area including a depression formed within the pocket, one end of the insert bit to be pushed into the depression to tip an opposite end of the insert bit out of the pocket.
 15. The multi-tool of claim 13, the storage area including an opening formed through the one of the handles within the pocket, the opening to provide access to the insert bit from an opposite side of the one of the handles.
 16. The multi-tool of claim 13, the storage area including a magnetic element to retain the insert bit within the pocket.
 17. The multi-tool of claim 11, the insert bit comprising a plurality of insert bits, and the storage area including a plurality of pockets to receive and store the plurality of insert bits.
 18. The multi-tool of claim 11, the one of the handles including another storage area to receive and store a chain link for a roller chain, in the closed position of the handles, the other of the handles to cover the another storage area.
 19. The multi-tool of claim 11, the one of the handles including another storage area to receive and store a wrench, the storage area formed in one side of the one of the handles and the another storage area formed in an opposite side of the one of the handles.
 20. The multi-tool of claim 11, the other of the handles including another storage area to receive and store a rasp, the another storage area including a recessed region formed in the other of the handles to receive and store the rasp.
-